Work with Delta Lake tables in Microsoft Fabric

1. Introduction to Microsoft Fabric

- **Unified Analytics Platform**: Microsoft Fabric is a centralized, software-as-a-service (SaaS) platform that integrates data engineering, data integration, warehousing, real-time intelligence, data science, and business intelligence into a single environment.
- **OneLake Storage**: It uses a centralized data storage architecture called OneLake, which unifies data across regions and clouds into a single logical lake. All compute engines in Fabric automatically store their data in OneLake.
- **Workload-Specific Tools**: Fabric includes several core services:
  - **Data Factory**: Used for ingesting, transforming, and orchestrating data workflows.
  - **Data Engineering**: Focuses on building, transforming, and sharing data estates through lakehouses.
  - **Data Science**: Provides tools to build, test, and deploy machine learning models.
  - **Power BI**: Enables the creation of interactive reports and dashboards for data-driven decisions.

2. Lakehouse Architecture and Collaboration

- **Hybrid Model**: A lakehouse in Fabric combines the flexible and scalable storage of a data lake with the querying and analytical capabilities of a relational data warehouse.
- **Role-Based Access**: Access is managed through workspaces using four main roles: admin, contributor, member, and viewer. Workspaces serve as logical containers to organize and manage data and reports.
- **Collaboration**: Fabric removes data silos, allowing data engineers, analysts, and scientists to work together in a single environment without duplicating efforts or data.

3. Working with Delta Lake Tables

- **Delta Lake Foundation**: Tables in a Fabric lakehouse are based on the open-source Delta Lake format, which adds relational database semantics to data lake storage.
- **Key Features of Delta Tables**:
  - **ACID Transactions**: Supports atomicity, consistency, isolation, and durability for reliable data modifications.
  - **Time Travel**: Transaction logs allow users to track row-level changes and retrieve previous versions of data.
  - **Schema Enforcement**: Ensures that data matches the defined schema to maintain data quality.
- **Table Management in Spark**: Data professionals can use Apache Spark to create "managed" tables (where Fabric manages both metadata and files) or "external" tables (mapped to an alternative storage location).

4. Optimization and Performance

- **Small File Problem**: Updates and deletes can result in many small files, slowing down queries.
- **Optimization Functions**:
  - **OptimizeWrite**: Reduces the number of files as they are written by creating fewer larger files.
  - **OPTIMIZE Command**: A maintenance tool that consolidates small Parquet files into larger ones for better performance.
  - **V-Order**: An optional sorting and encoding technique that enables "lightning-fast" reads in Fabric compute engines.
  - **VACUUM**: Permanently removes old data files that are no longer referenced in the transaction log and fall outside the retention period.

5. Streaming and AI Integration

- **Structured Streaming**: Spark provides native support for streaming data, allowing Delta tables to serve as either a source (for near real-time reporting) or a sink (for capturing data from IoT devices).
- **Copilot Productivity**: Microsoft Fabric integrates Copilot, a generative AI assistant that provides intelligent code completion, automated routine tasks, and natural language-to-SQL conversion across all workloads.

Questions

1. Which of the following descriptions best fits Delta Lake?
A. A Spark API for exporting data from a relational database into CSV files.
B. A relational storage layer for Spark that supports tables based on Parquet files.
C. A synchronization solution that replicates data between SQL Server and Spark.
D. A cloud-native hardware acceleration for Spark processing.

2. You've loaded a Spark dataframe with data, that you now want to use in a delta table. What format should you use to write the dataframe to storage?
A. CSV
B. PARQUET
C. DELTA
D. JSON

3. You have a managed table based on a folder that contains data files in delta format. If you drop the table, what happens?
A. The table metadata and data files are deleted.
B. The table definition is removed from the metastore, but the data files remain intact.
C. The table definition remains in the metastore, but the data files are deleted.
D. The table is automatically converted into an external table.

4. Tables in a Microsoft Fabric lakehouse are based on which open-source storage format?
A. Apache Avro
B. Delta Lake
C. Apache Hive
D. Protocol Buffers

5. Where does a Delta table store its transaction logs to ensure ACID compliance?
A. In a hidden system database.
B. In a _delta_log folder within the table's storage path.
C. In the Spark driver's local memory.
D. In a separate SQL Server instance.

6. What is the primary purpose of the OptimizeWrite feature in Microsoft Fabric?
A. To encrypt data for enhanced security.
B. To reduce the number of small files to prevent performance degradation.
C. To automatically partition data by the current year.
D. To compress CSV files into ZIP format.

7. Which command should be used to permanently remove old Parquet data files that are no longer referenced in the transaction log?
A. OPTIMIZE
B. CLEANUP
C. VACUUM
D. PURGE

8. What is the default retention period for data files when running a VACUUM command in Fabric?
A. 24 hours
B. 7 days
C. 30 days
D. 48 hours

9. Which Microsoft Fabric optimization provides lightning-fast reads by applying special sorting and row group distribution?
A. Z-Order
B. V-Order
C. Delta-Speed
D. Verti-Scan

10. When using an external table, what happens to the underlying data files if the table is deleted from the metastore?
A. The data files are deleted immediately.
B. The data files remain in their storage location.
C. The data files are moved to a temporary trash folder.
D. The data files are automatically converted to CSV format.

11. Which feature allows a user to query an older version of data by referencing the transaction log?
A. Time Travel
B. Data Rollback
C. Version Snapshotting
D. Transaction Replay

12. In Spark Structured Streaming, what is the purpose of the checkpointLocation option?
A. To define the schema of the incoming stream.
B. To track the state of stream processing for recovery from failure.
C. To limit the number of files written per second.
D. To specify the backup location for Parquet files.

13. Under what condition can partitioning a table actually degrade performance?
A. When you have very large amounts of data.
B. When tables are split into a few large partitions.
C. When data volumes are small or a partitioning column has high cardinality.
D. When the table is stored in the "Tables" area of the lakehouse.

Answers

1. B
2. C
3. A
4. B
5. B
6. B
7. C
8. B
9. B
10. B
11. A
12. B
13. C